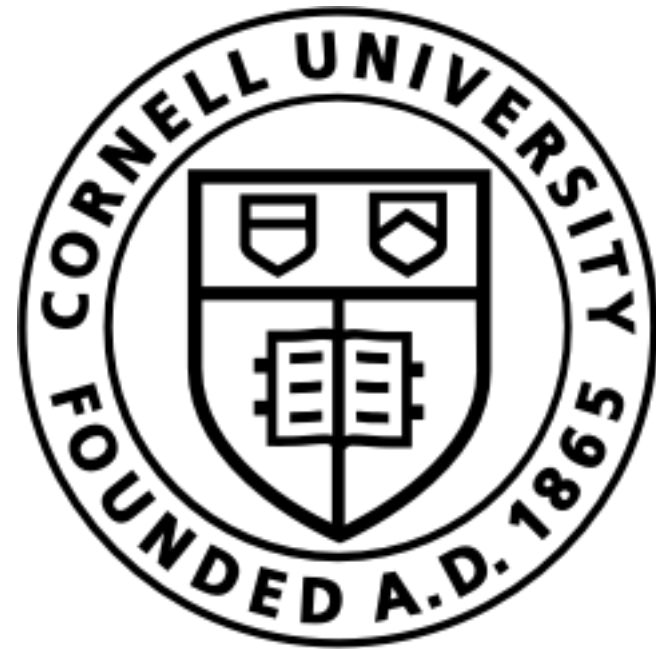




Lecture 6a: Revisiting the Certainty Equivalent and Risk Aversion, and an Introduction to a Binomial Future

CHEME 5760 Cornell University

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Training only. This content is for training and informational purposes only. The teaching team does not make, give, or endorse any offer or solicitation to buy or sell securities or derivative products or provide investment or trading advice or strategy.

Risk. Trading involves risk and is generally not recommended for those with limited resources, experience, or a low-risk tolerance. Past performance does not guarantee future success.

Responsibility. You are responsible for your investment decisions based on your financial circumstances, objectives, risk tolerance, and liquidity needs.

Objectives. In this lecture, we will:

- ▶ **O1:** Introduce the Midpoint Certainty Equivalent Method (MCEM) to compute your Certainty Equivalent independently from risk lottery choices
- ▶ **O2:** Revisit the Certainty Equivalent (CE), the Risk Premium (RP), and Risk Aversion (RA) in the context of the risky+risk-free investment problem.
- ▶ **O3:** Introduce the Arrow-Pratt Risk Aversion Measure as a means to bound our risk aversion parameter for an arbitrary utility function

———— next time ————

Introduction to a binomial future. We develop a discrete model to make decisions over many possible uncertain states

Jupyter Notebooks

Repository: <https://github.com/varnerlab/CHEME-5760-Examples-F23.git>

- **Example:** CHEME-5760-L6a-ArrowPratt.ipynb

O1: Introduce the Midpoint Certainty Equivalent Method (MCEM) to compute your Certainty Equivalent independently from risk lottery choices

Reference: John C. Hershey and Paul J. H. Schoemaker (1985). Probability Versus Certainty Equivalence Methods in Utility Measurement: Are they Equivalent? Manage. Sci. v31, issue 10, 1213–1231. <https://doi.org/10.1287/mnsc.31.10.1213>

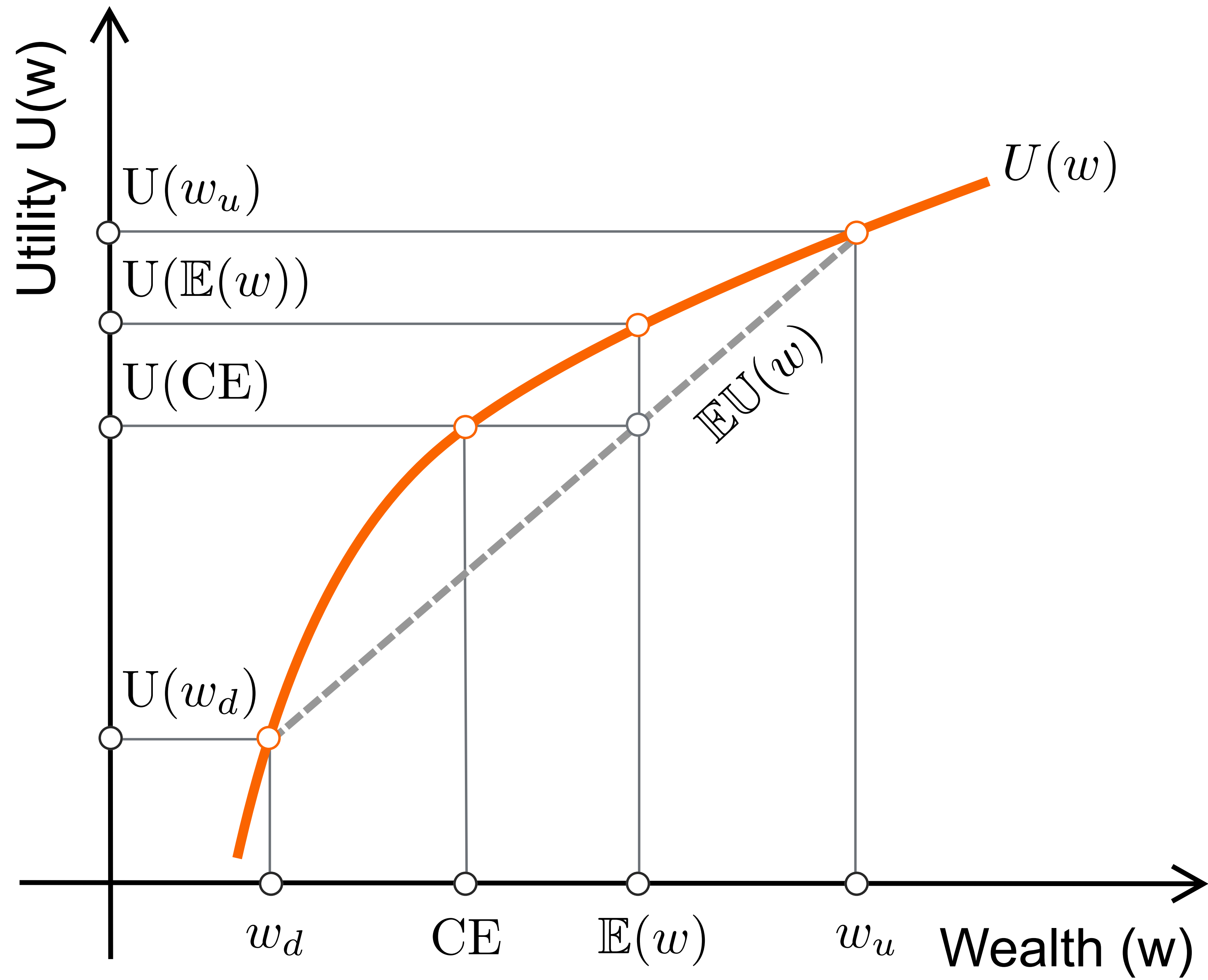
O2: Revisit the Certainty Equivalent (CE), the Risk Premium (RP), and Risk Aversion (RA) in the context of the risky+risk-free investment problem.

An investor's utility function is $U(w, \tau) = w^\tau$, where w is the investor's wealth and τ is the investor's risk aversion parameter. The certainty equivalent CE of a risky lottery is described by:

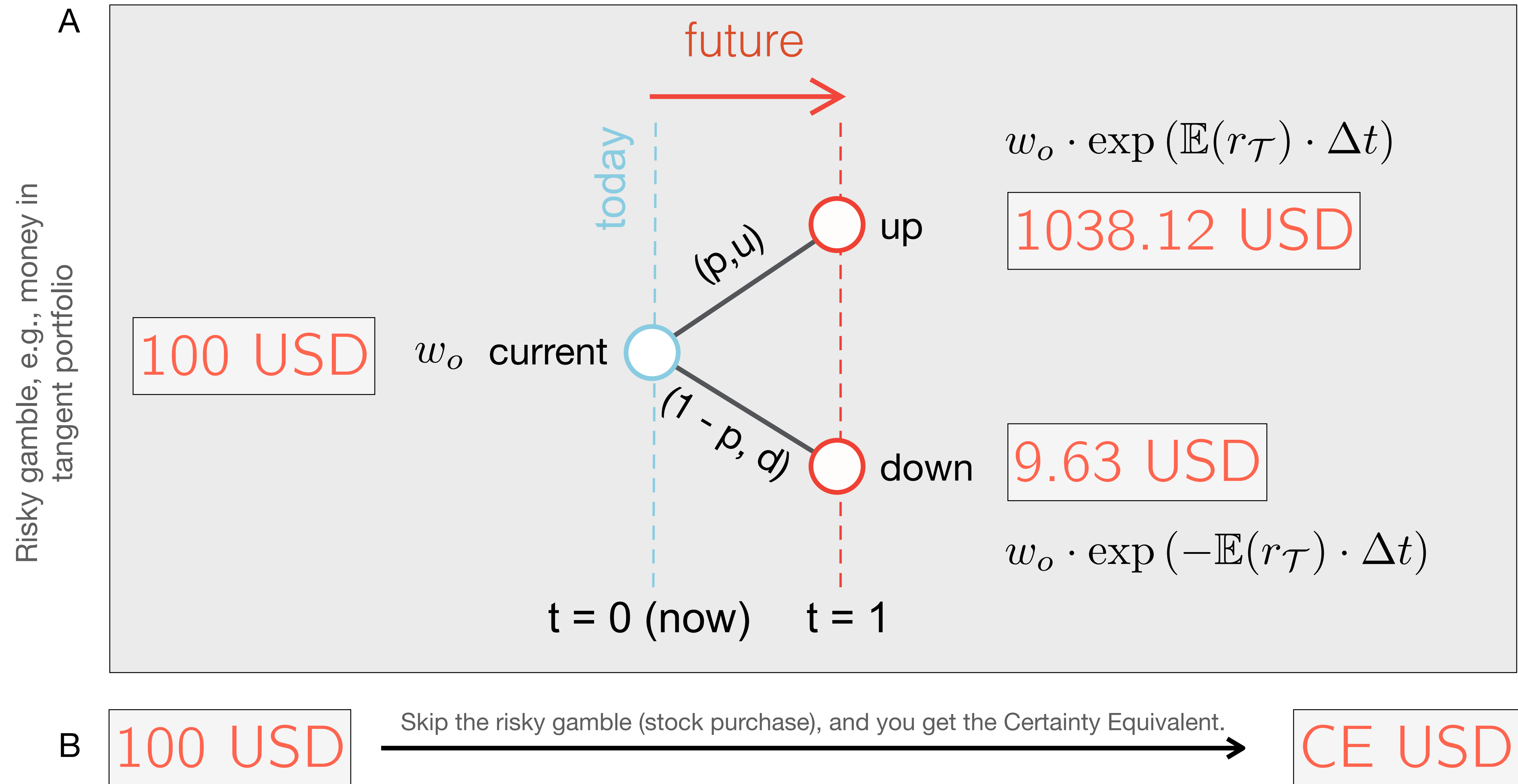
$$U(CE) = p \cdot U(w_u) + (1 - p) \cdot U(w_d)$$

where w_u is the wealth of a favorable outcome (up state), w_d is the wealth of an unfavorable outcome (down state), and p is the probability of the favorable outcome. If the investor specifies the certainty equivalent CE, we can solve for risk aversion parameter τ :

$$\tau = \frac{\ln\left(p \cdot U(w_u) + (1 - p) \cdot U(w_d)\right)}{\ln(CE)}$$



Concept: Estimate Your Certainty Equivalent (CE)



O3: Introduce the Arrow-Pratt Risk Aversion Measure as a means to bound our risk aversion parameter for an arbitrary utility function

Let a decision maker have a twice differentiable utility function $U : X \rightarrow \mathbb{R}$, where X is a set of outcomes or states involved in a decision, e.g., wealth, consumption, etc. The Arrow-Pratt measure of absolute risk aversion is defined as:

$$r(x) = -\frac{U''(x)}{U'(x)}$$

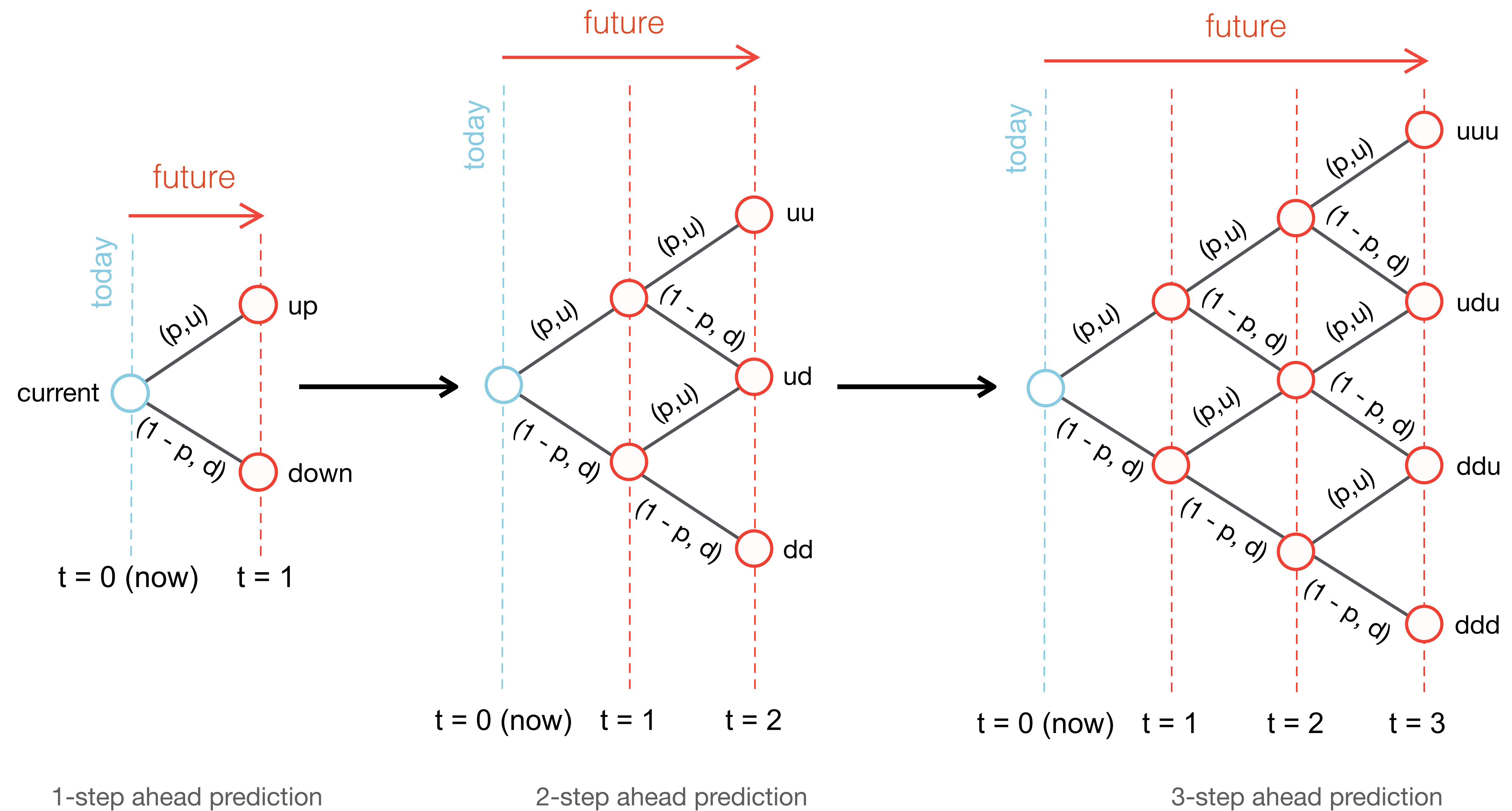
Values of the Arrow-Pratt aversion measure $r(x)$ can be used to classify the risk attitude of a decision maker:

- If $r(x) > 0$ for values of x , then the decision maker is **risk averse**.
- If $r(x) = 0$ for values of x , then the decision maker is **risk neutral**.
- If $r(x) < 0$ for values of x , then the decision maker is **risk loving**.

Pratt, J. W. (1964). Risk Aversion in the Small and in the Large. *Econometrica*, 32(1/2), 122–136. <https://doi.org/10.2307/1913738>

Example: CHEME-5760-L6a-ArrowPratt.ipynb

Next time: Introduction to a binomial future. Let's develop a discrete model to make decisions over many possible uncertain states



Today?